VICTORYHEALTH ANALYTICS

MedSafe-R

Reliability, maintenance and patient-safety analysis

Corrected synthetic dataset v1.3 | snapshot 31 July 2026

Overdue PM: 253 devices (21.1%); log-only evidence gives 491 overdue among 1161 devices with a recorded PM, with 39 unknown.

Corrective repairs account for 76.3% of recorded maintenance downtime. 130 devices have at least 3 recorded repairs.

Failure in the next 30 days: logistic AUC 0.608; random forest 0.679; heuristic 0.703.

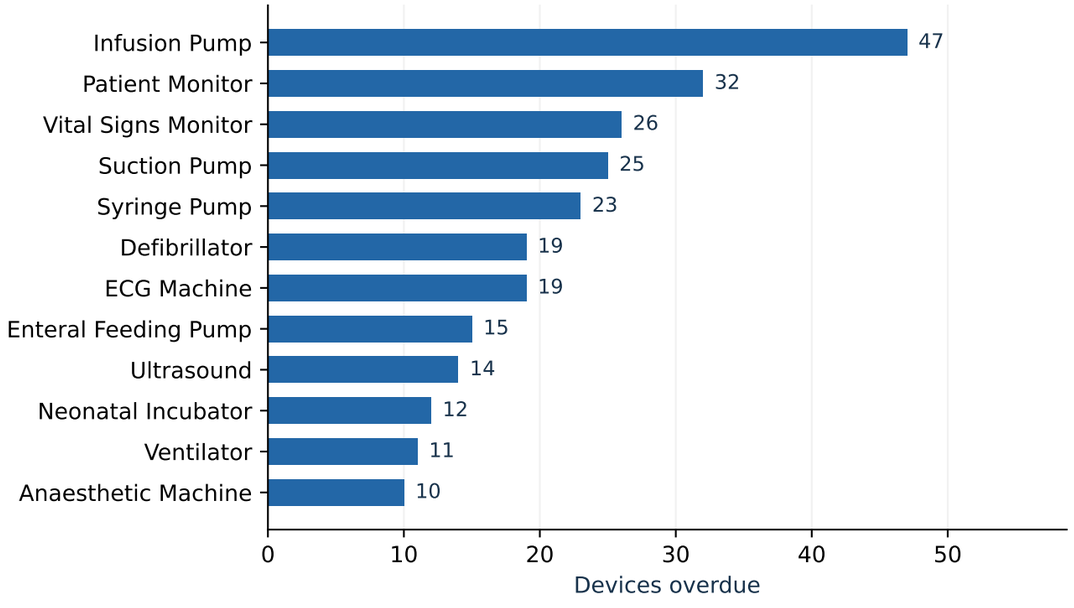
Safety event in the next 90 days: logistic AUC 0.557; random forest 0.582; heuristic 0.619.

Five analysis themes, six chart sheets, auditable tables and reproducible baseline models.
Prepared for Victoria Ekweani. Exploratory portfolio prototype; not a clinical decision tool.

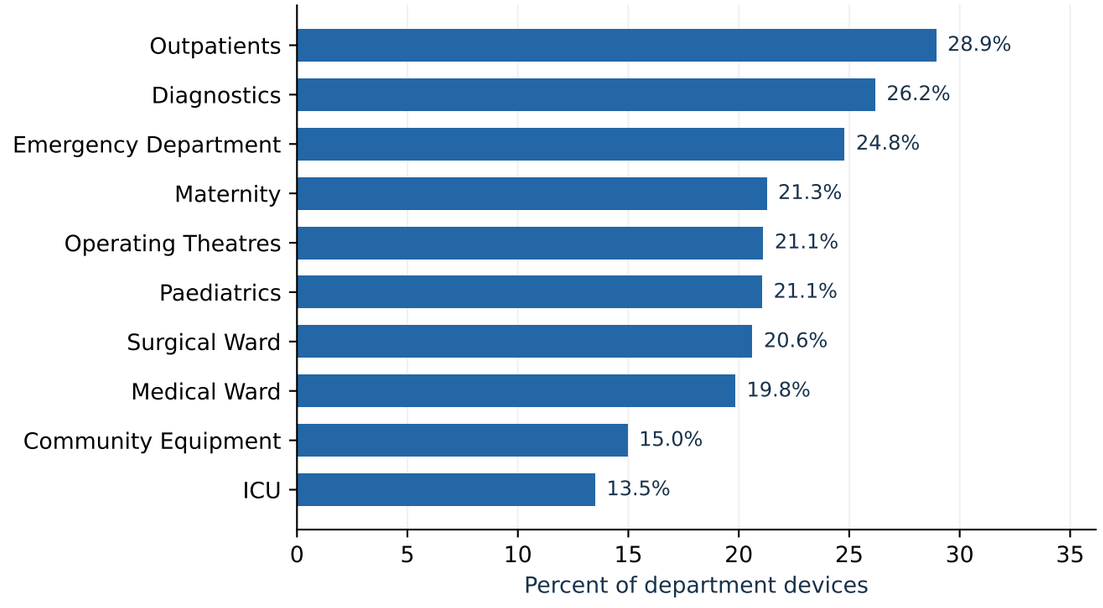
01 | Overdue preventive maintenance

253 / 1,200 devices overdue (21.1%) at 31 July 2026. Inventory-based estimate; PM source assumptions apply.

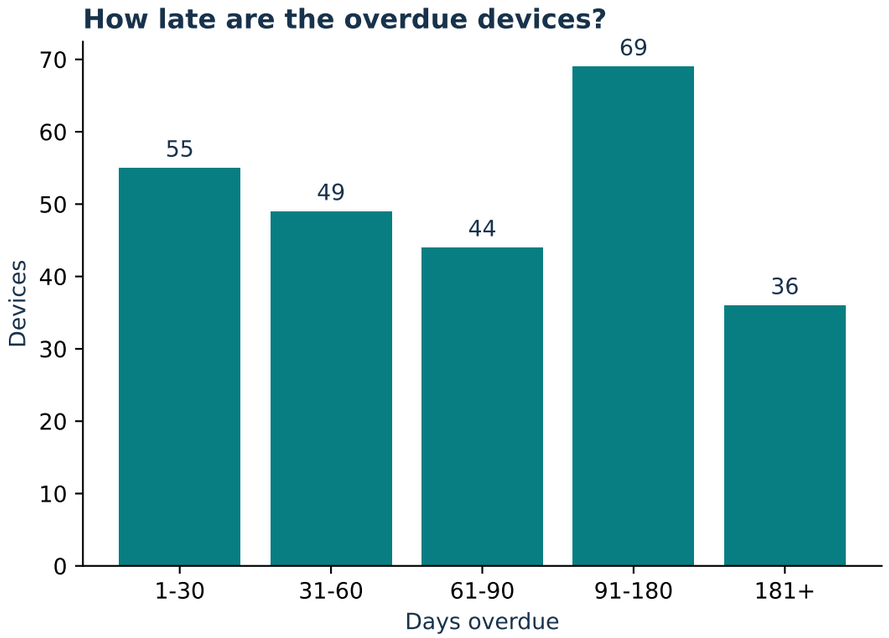
Overdue workload by device type



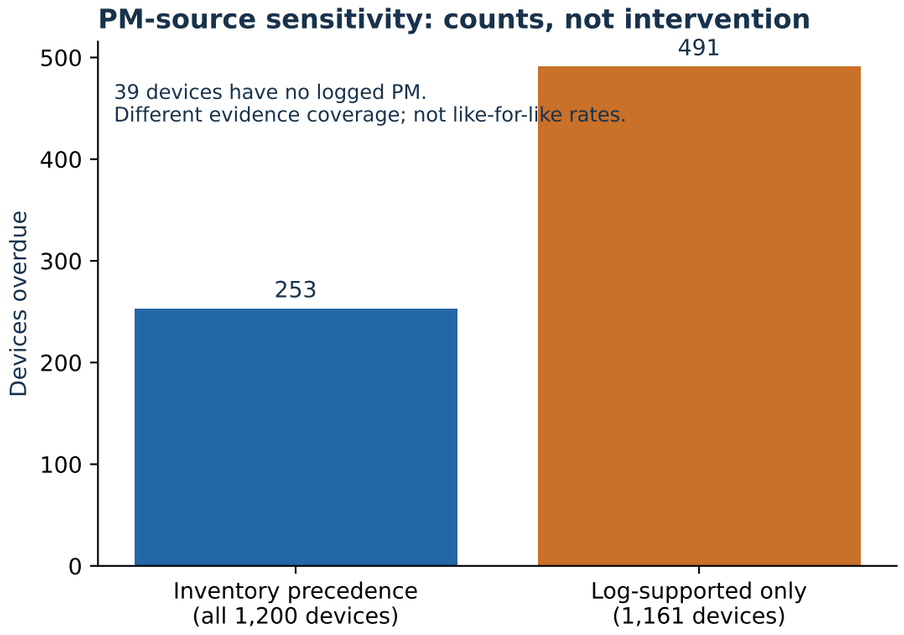
Overdue share by department



How late are the overdue devices?



PM-source sensitivity: counts, not intervention

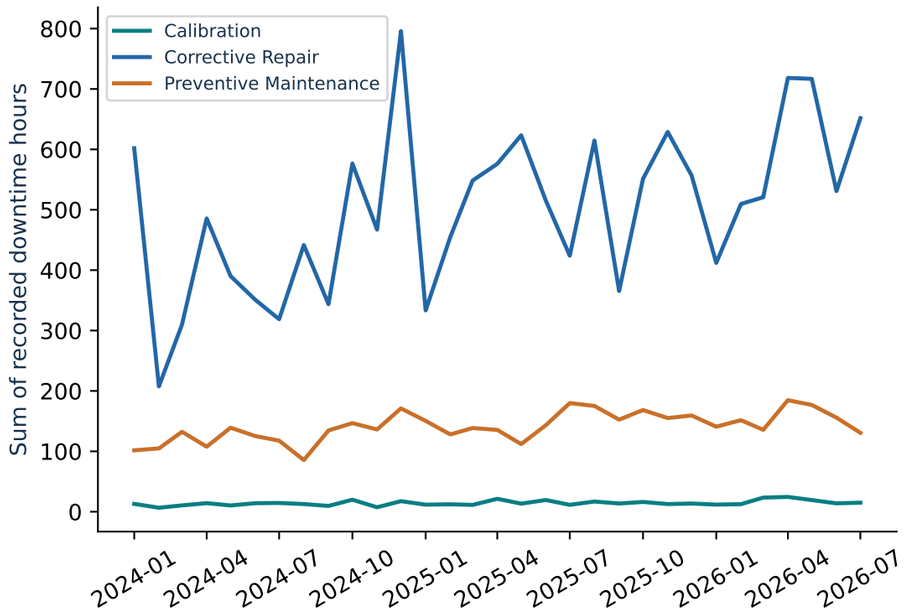


Synthetic data. PM precedence: later of inventory and valid PM log. 639 inventory-only dates; 11 commissioning placeholders.
Full overdue-device register and source breakdown are supplied as CSV tables.

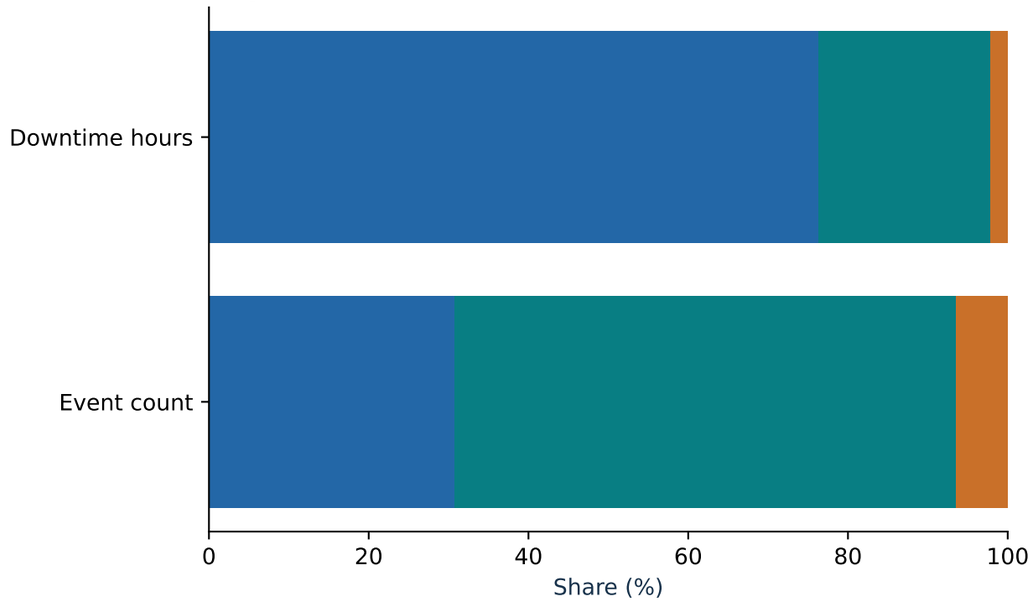
02 | Downtime and recurring repairs

1,341 corrective repairs; 125 occurred within 90 days of the previous corrective repair on the same device.

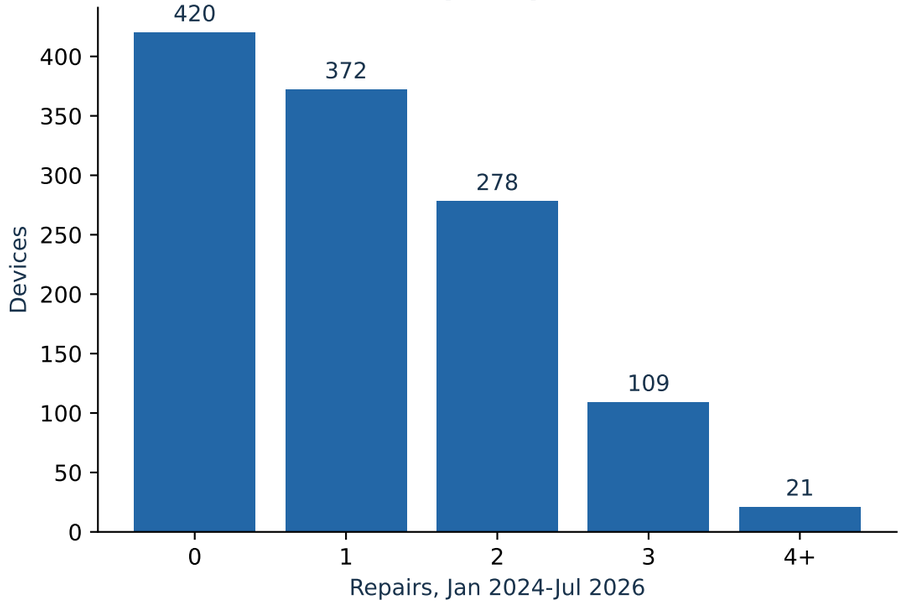
Maintenance-associated downtime by month



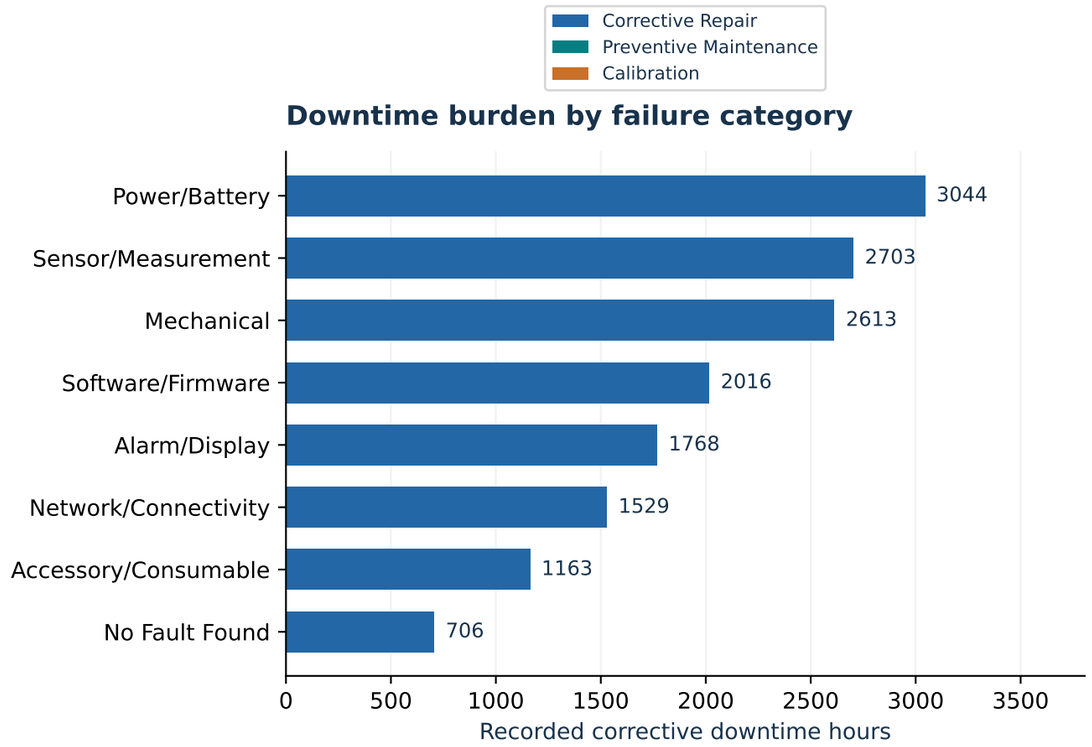
Repair workload versus downtime burden



Recorded corrective repairs per device



Downtime burden by failure category

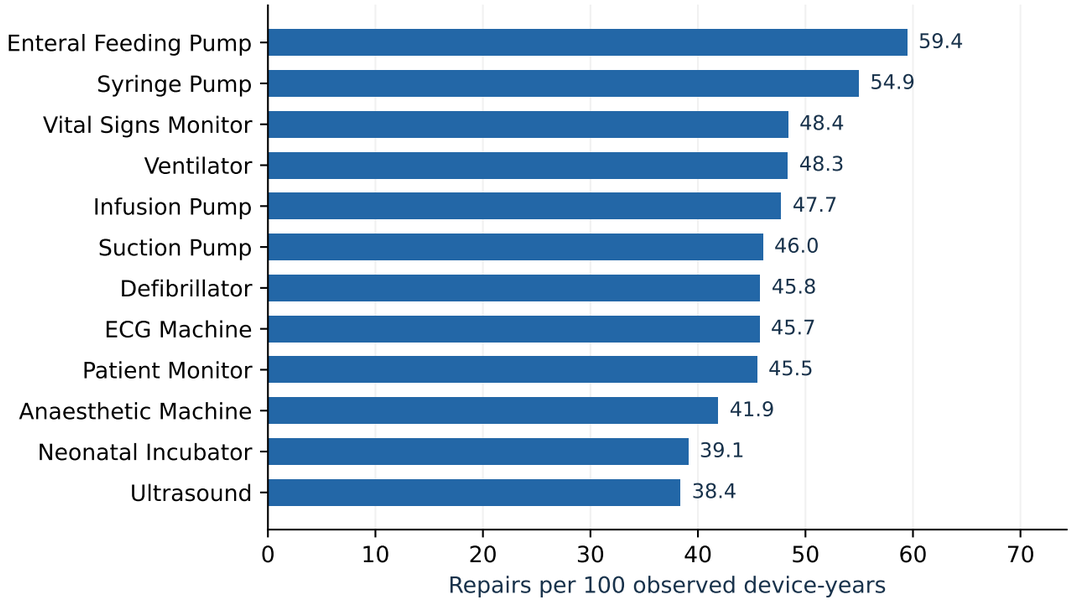


Repeat = another corrective repair within 90 calendar days, regardless of category; not proof of the same unresolved fault.
Recorded downtime sums are event burden, not validated fleet unavailability: event durations may overlap.

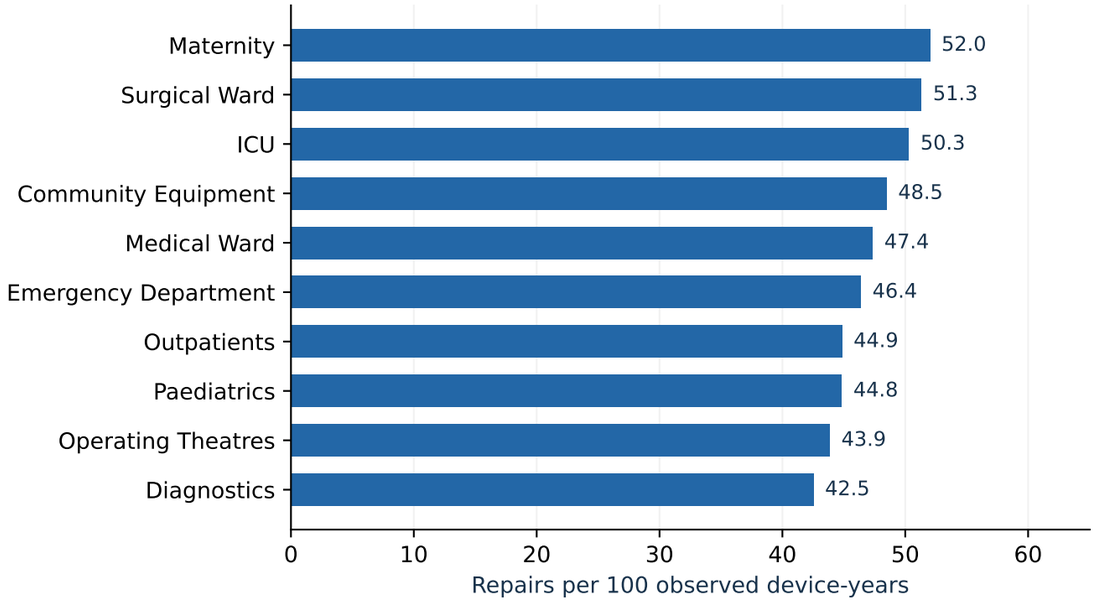
03 | Maintenance needs: compare burden and rates

Repair rates use calendar device-years since commissioning or 1 Jan 2024, whichever is later; they are not operating-hour failure rates.

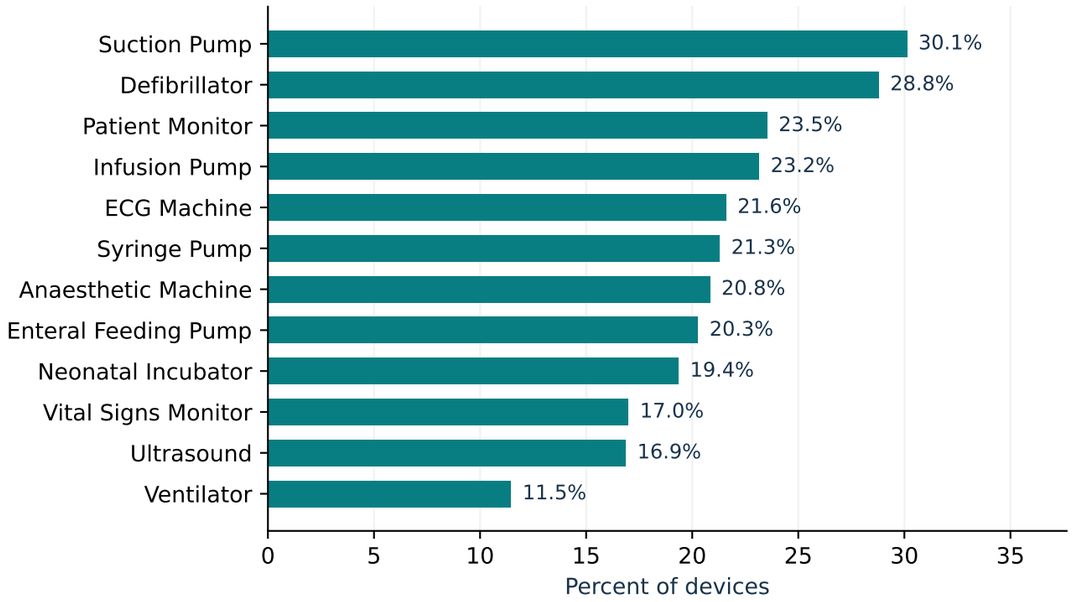
Corrective repairs by device type



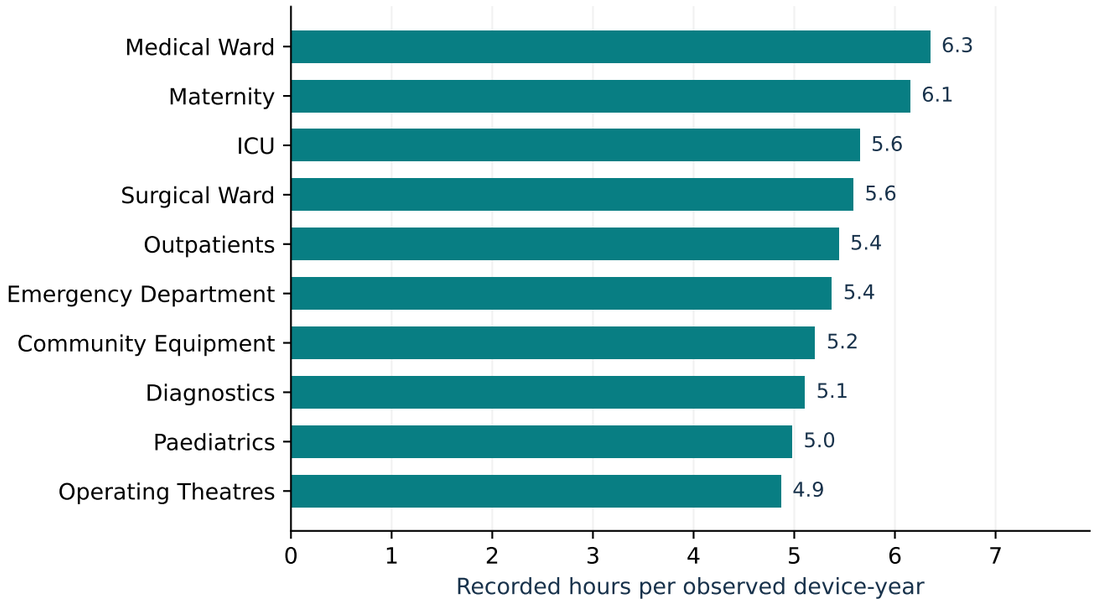
Corrective repairs by department



Overdue PM share by device type



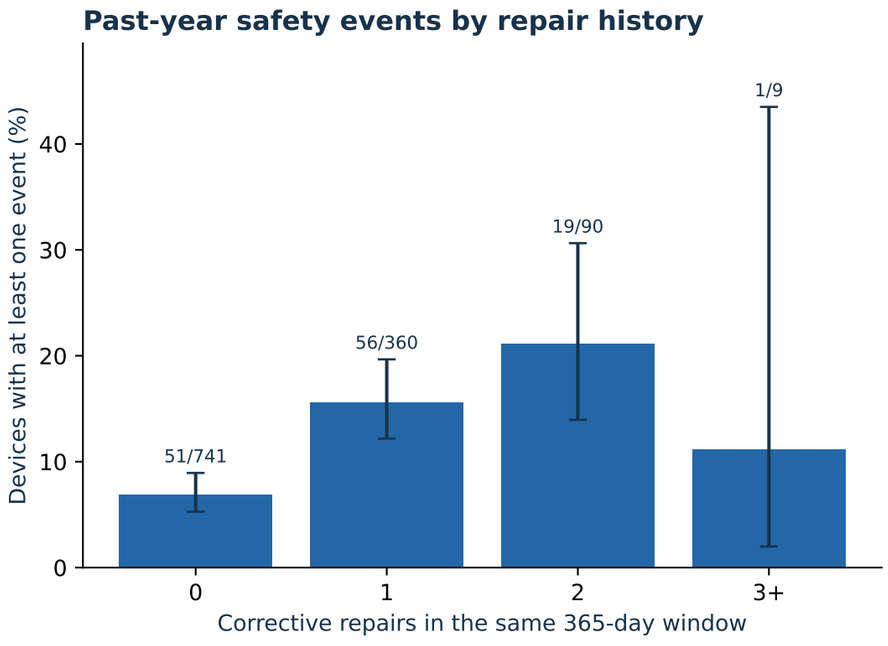
Repair downtime burden by department

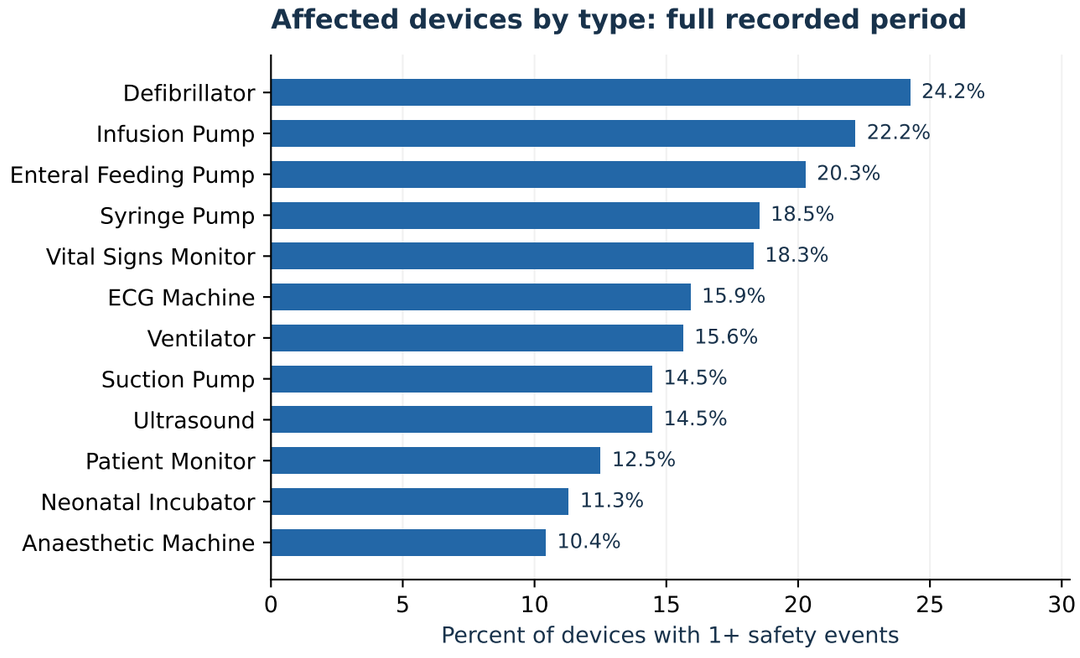


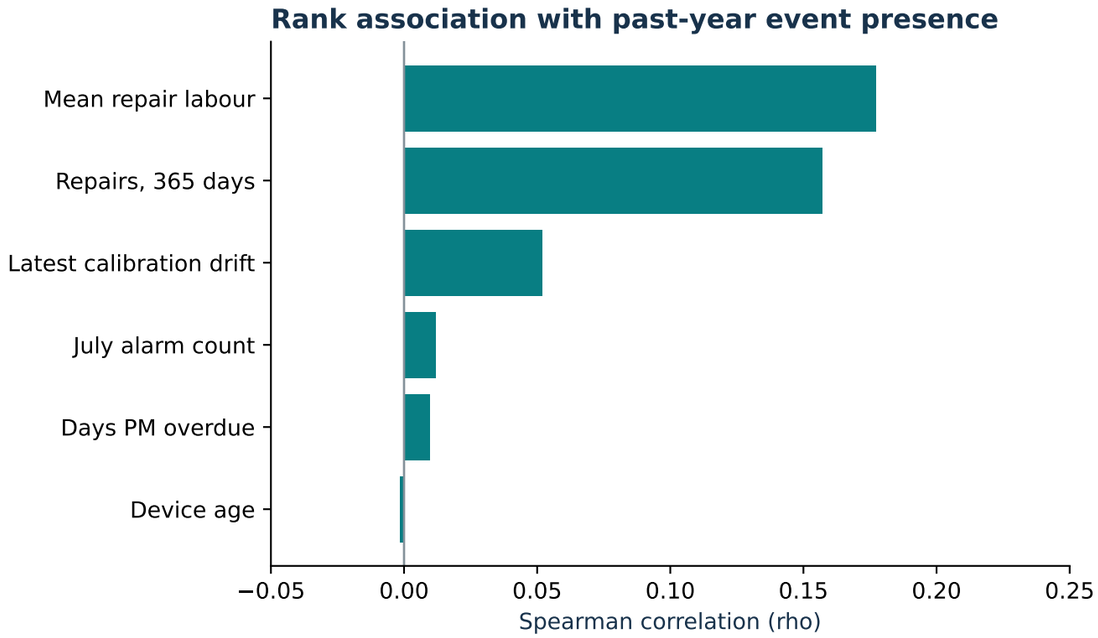
Descriptive comparisons. No claim of causal or statistically established differences; age, type mix, utilisation and logging differ.
Time-at-risk assumes continuous observation after commissioning within the study; decommissioning dates are unavailable.

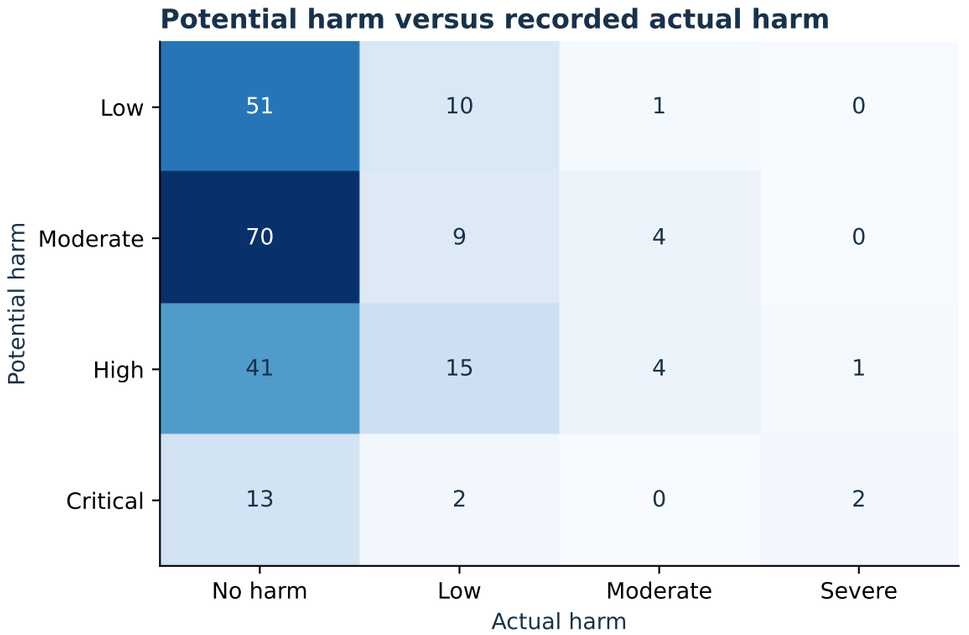
04 | Reliability indicators and simulated safety events

223 recorded events affect 206 distinct devices. Associations below do not establish causality or future prediction.





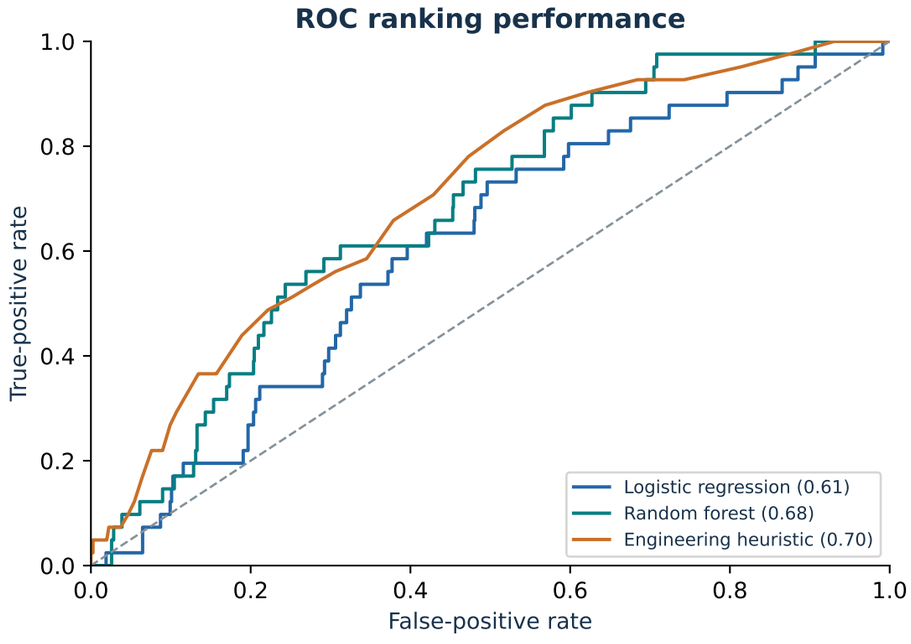


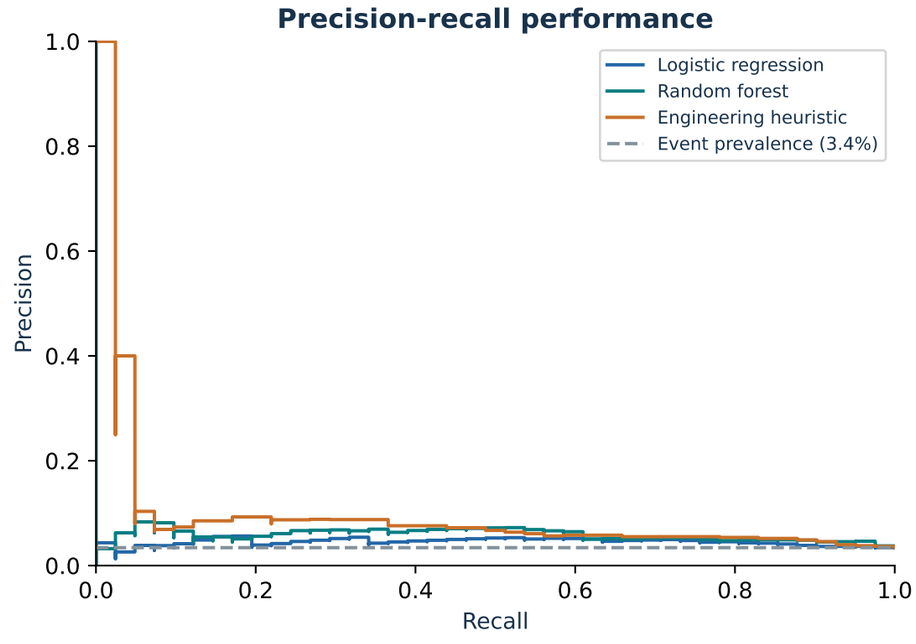


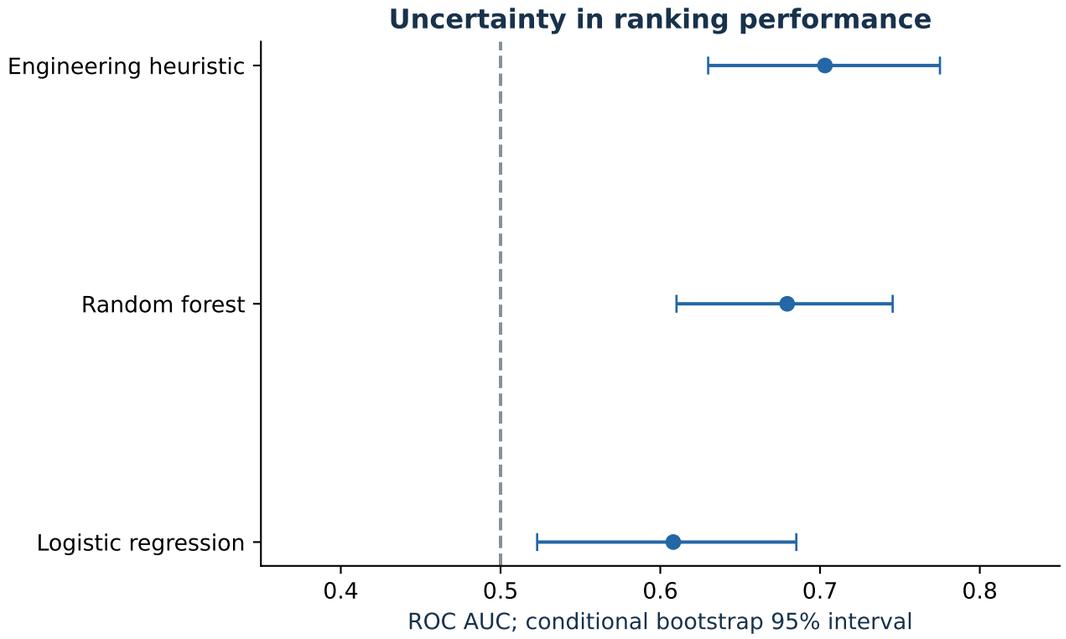
Error bars: Wilson 95% intervals, descriptive of the synthetic sample only. Small groups are unstable.
Past-year window: 1 Aug 2025-31 Jul 2026. Same-window associations may include repairs following safety events; no temporal causal ordering.

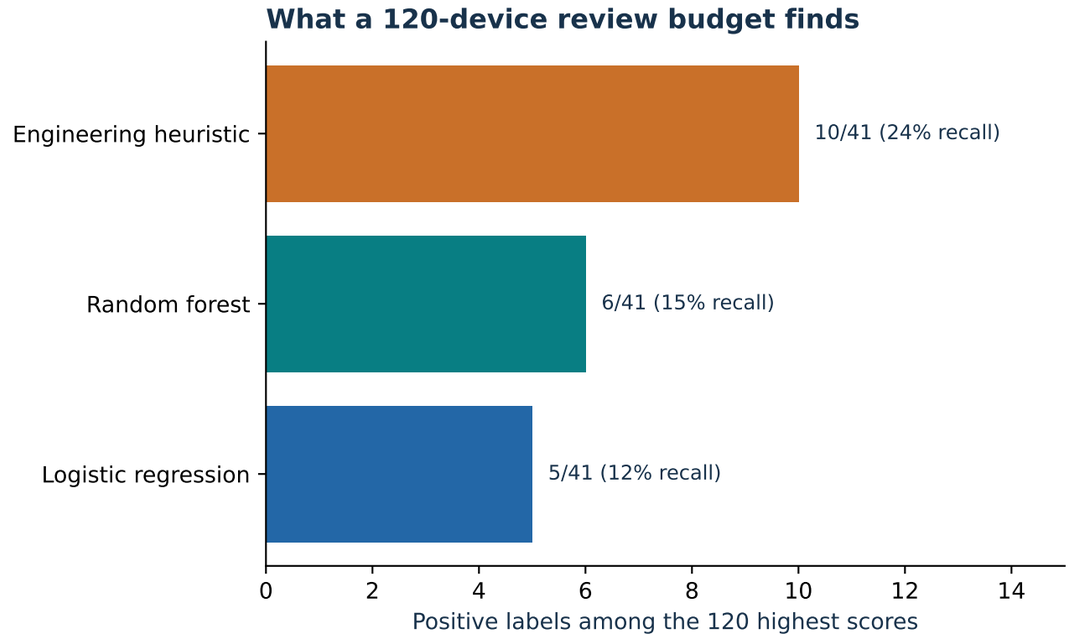
05 | Failure in the next 30 days - synthetic-label baseline

41 positive labels / 1,200 devices (3.42%). Fixed model settings; stratified 5-fold CV repeated 3 times; averaged out-of-fold scores.





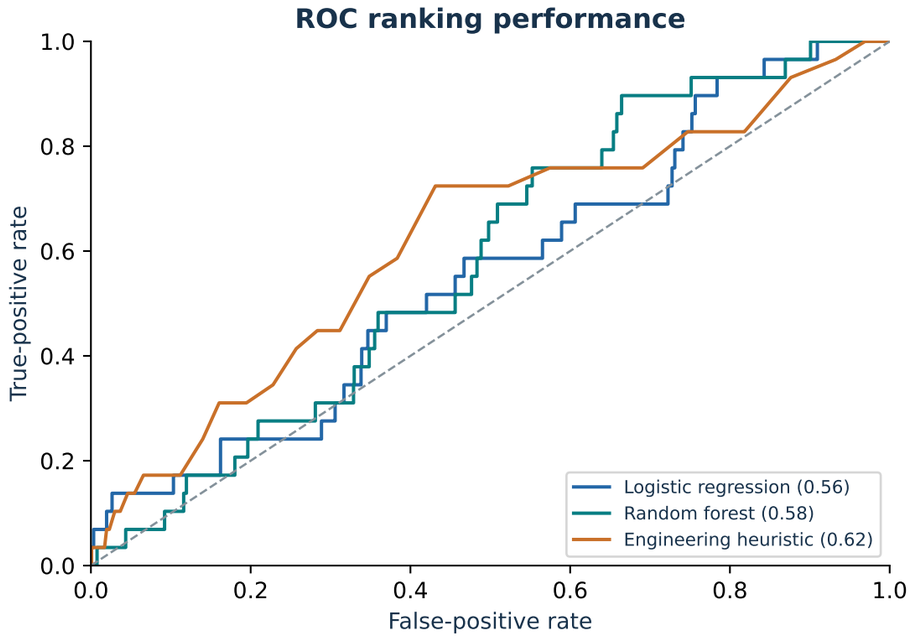


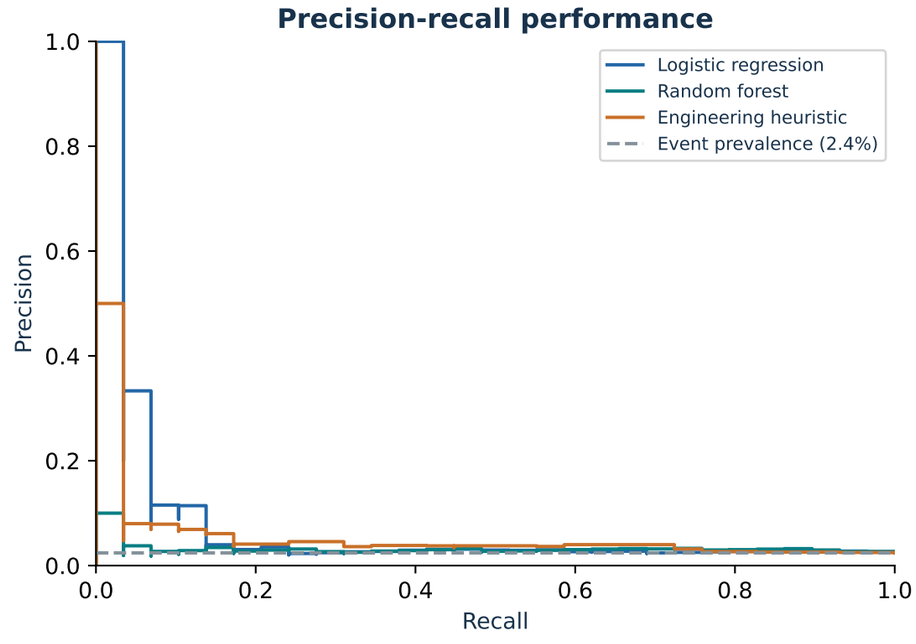


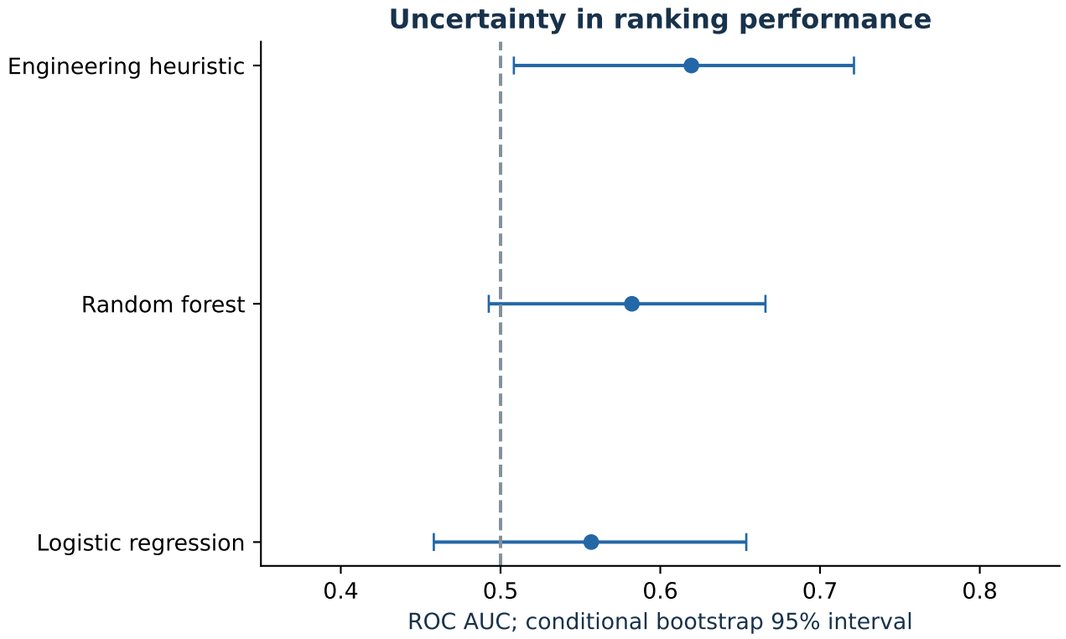
Intervals resample devices with fixed out-of-fold scores; they do not include retraining or generator uncertainty. No independent external test.
Synthetic labels lack future event verification. Heuristic is a ranking score, not a probability; shared generator assumptions may favour it.

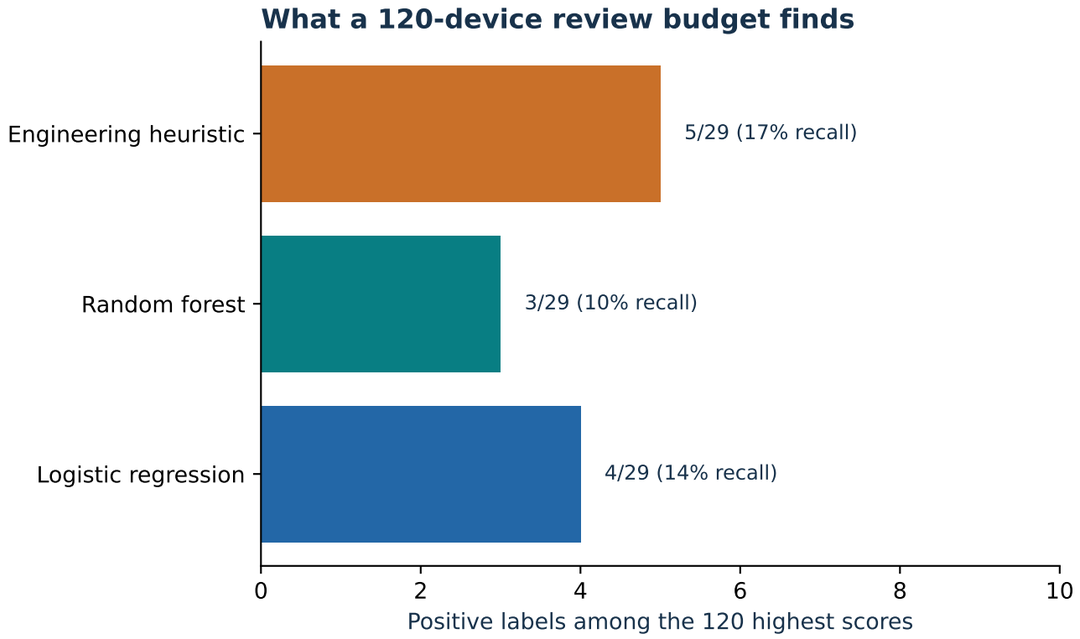
05 | Safety event in the next 90 days - synthetic-label baseline

29 positive labels / 1,200 devices (2.42%). Fixed model settings; stratified 5-fold CV repeated 3 times; averaged out-of-fold scores.









Intervals resample devices with fixed out-of-fold scores; they do not include retraining or generator uncertainty. No independent external test.
Synthetic labels lack future event verification. Heuristic is a ranking score, not a probability; shared generator assumptions may favour it.

Methods, assumptions and limitations

1. All records, manufacturers, regulatory signals and outcomes are synthetic. Results show analytical methods, not validated device risk or patient-safety benefits.

2. Snapshot: 31 July 2026. Historical event windows are exactly 90 days (3 May-31 July 2026) and 365 days (1 August 2025-31 July 2026), inclusive. July telemetry is a calendar month; utilisation is the May-July mean.

3. Overdue PM uses the later of inventory and valid logged PM. Source coverage: 550 log-supported, 639 inventory-only, 11 commissioning placeholders. The latter do not prove PM occurred. Sensitivity counts use a different evidence denominator.

4. Nine monthly records fail hours plausibility checks. They are excluded from hours/utilisation analysis; none is in May-July. Reported downtime uses maintenance logs, not monthly operations; missing commissioning-month exposure remains an assumption.

5. Corrective repairs are not guaranteed independent failures. Repeat jobs within 90 days may be unrelated categories. Mean repair intervals are calendar gaps, not operating-hour MTBF; legacy 999 values are excluded from modelling.

6. Safety affected-device counts use distinct device IDs. Historical associations use binary presence of an event in the same 365-day period and do not establish event ordering or causality. Wilson intervals are descriptive only.

7. Learning pipelines fit preprocessing only on training folds. Models have fixed settings, no hyperparameter search. No legacy MTBF, ID, priority score/band, synthetic quality score or target enters the trained predictors. Missing repair intervals are imputed within folds with indicators.

8. Cross-validation assesses synthetic-label discrimination among held-out devices in the same dataset. It is not prospective, temporal or external validation. Some predictors and labels may share generator logic, and labels have not been checked against future event records.

9. Conditional bootstrap AUC intervals quantify resampling uncertainty for fixed OOF scores. They do not capture model refitting, correlated site effects or uncertainty in the data generator. Comparisons are descriptive, not significance tests.

10. Class weighting and resampling are not used; rare-event performance is reported with average precision, Brier score (trained probability models only), and positives found in a fixed top-120 review budget. No deployment threshold is validated. Calibration requires further study.

11. CSV tables, fold assignments/predictions, exact environment versions and source hash are included. Run python analysis.py after installing requirements.txt. Each HTML chart has a corresponding table view; chart images themselves are static.